HW 4 — Jackson Eq. 7.90

black = write it down · red = the answer · teal = my note, do not copy · grey box = optional

Problem — General solution of the 1D wave equation

> Derive **Jackson Eq. 7.90**, the general solution of the one-dimensional wave equation.

What to write: Steps 0 → 3, ending at Eq. 7.90. That is the whole answer. Everything after the dashed line on page 4 is background, not required.

ANSWER

$$u(x,t) = \frac{1}{2}\,\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dk\; A(k)\, e^{i[kx - \omega(k)t]} + \text{c.c.}$$

*c.c. = complex conjugate of the term before it. A(k) is a **free function** — that is what makes the solution general.*

Where each step comes from in the question:

"wave equation"	Step 0
"one-dimensional" → $\nabla^2 \to \partial^2/\partial x^2$	Step 1
"**general**" → all k, not one wave	Step 2
u is physical → real → the "+ c.c."	Step 3

Step 0 — Starting point

Maxwell in vacuum gives the wave equation:

$$\nabla^2 \vec{E} - \mu_0\varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} = 0, \qquad \mu_0\varepsilon_0 = \frac{1}{c^2}$$ Lec 1, slide 13

Each component obeys it separately. Take E_x, expand in a Fourier integral:

$$E_x(\vec{x},t) = \frac{1}{(2\pi)^{3/2}} \int d^3k\; e^{i\vec{k}\cdot\vec{x}}\; \varepsilon_x(\vec{k},t)$$ Lec 1, slide 16

*3D → 1D: $d^3k \to dk$, $(2\pi)^{3/2} \to \sqrt{2\pi}$. ε_x still depends on t — it is a(k,t) of Step 1, **not** Jackson's A(k). Result also stated in* Lec 8, slide 9

Step 1 — One dimension; time dependence of each k

Write $u = E_x$. Jackson uses the name u because every component obeys the same scalar equation — the derivation does not care which field it is.

In 1D, $\nabla^2 = \partial^2/\partial x^2$:

$$\frac{\partial^2 u}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = 0$$

Write u as a **Fourier integral**:

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int dk \; e^{ikx} a(k,t)$$

Here **a(k,t)** = how much of the wave e^{ikx} is present in u at time t.

Differentiate twice in **x** — only e^{ikx} depends on x, and each ∂/∂x brings down ik:

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{\sqrt{2\pi}} \int dk\, (ik)^2 e^{ikx} a = \frac{1}{\sqrt{2\pi}} \int dk\, (-k^2)\, e^{ikx} a$$

Differentiate twice in **t** — only a depends on t:

$$\frac{\partial^2 u}{\partial t^2} = \frac{1}{\sqrt{2\pi}} \int dk\, e^{ikx} \frac{\partial^2 a}{\partial t^2}$$

Put both into the wave equation:

$$\frac{1}{\sqrt{2\pi}} \int dk\, e^{ikx} \left[-k^2 a - \frac{1}{c^2}\frac{\partial^2 a}{\partial t^2} \right] = 0$$

This must hold for **every x**. A Fourier integral can only vanish for all x if its amplitude vanishes for **each k separately** — there is nothing left to cancel against. So the bracket is zero; multiplying it by $-c^2$:

$$\frac{\partial^2 a}{\partial t^2} + k^2 c^2 a = 0 \quad \text{(Helmholtz)}$$ Lec 1, slide 18

What this achieved. *u(x,t) depended on two variables and obeyed a partial differential equation. a(k,t) depends on t only — k is just a label — so it obeys an* ***ordinary*** *differential equation, one for each k. That is the whole point of the Fourier integral.*

Its two solutions are $e^{-i\omega t}$ and $e^{+i\omega t}$:

$$a(k,t) = C(k)\, e^{-i\omega t} + D(k)\, e^{+i\omega t}, \qquad \omega = c|k| \quad \text{Lec 1, slide 18}$$

C and D are the two constants of integration, fixed in Step 3.

Slide 18 writes $\omega = kc$, where $k = |\vec{k}|$ is a 3D magnitude, never negative. In 1D k is signed, so $\omega = c|k|$ — even in k, as needed next.

$\omega = c|k|$ is the **vacuum** answer. In glass or water the speed depends on frequency, so the curve is different. Jackson therefore writes $\omega = \omega(k)$, meaning "whatever function of k it happens to be", so that 7.90 holds in any medium:

$$\omega = \omega(k), \qquad \omega(-k) = \omega(k) \qquad \text{Jackson 7.79}$$

The second condition says a wave going left has the same frequency as the same wave going right. It is used in Step 3, where k is flipped to -k.

Step 2 — Superposition over all k

> **The method in three moves:** Step 1 first **broke** u into one piece per k, then **solved** each piece. Step 2 **glues** the pieces back together.

Put a(k,t) of Step 1 back into the Fourier integral of Step 1:

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int dk \; e^{ikx} \left[C(k)\, e^{-i\omega t} + D(k)\, e^{+i\omega t} \right]$$

Combine the exponentials in each term, $e^{ikx}e^{\mp i\omega t} = e^{i(kx \mp \omega t)}$:

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk \; \left[C(k)\, e^{i(kx - \omega t)} + D(k)\, e^{i(kx + \omega t)} \right]$$

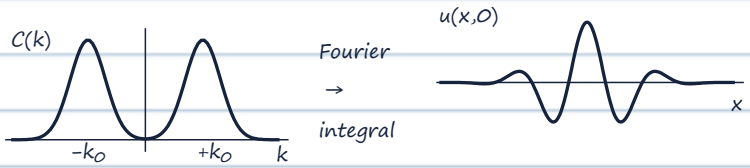
Both branches are kept. *Nothing is discarded here — Step 3 will show that D is not free, it is fixed by C.*

This is the **general** solution because:

① the equation is **linear** → any sum of solutions is a solution;

② C(k) and D(k) are free for every k.

One k = an endless sine wave. A band of k added together = a pulse:



Narrow band in k → long pulse in x; wide band → short pulse.

Step 3 — Make u real → Eq. 7.90

u is a physical field, so it must be real: $u = u^*$.

Conjugate the whole expression:

$$u^* = \frac{1}{\sqrt{2\pi}} \int dk \; e^{-ikx} \left[C^*(k)\, e^{+i\omega t} + D^*(k)\, e^{-i\omega t} \right]$$

Why relabel. Compare the two lines: u carries e^{+ikx}, but u^* carries e^{-ikx}. Different x-dependence, so the coefficients cannot be matched yet. Substituting $k \to -k$ in the u^* integral flips it:

	before	after $k \to -k$
the x-factor	e^{-ikx}	e^{+ikx} ✓
the amplitude	$C^*(k)$	$C^*(-k)$
the frequency	$\omega(k)$	$\omega(-k) = \omega(k)$
the limits	$-\infty \to \infty$	$\infty \to -\infty$, and $dk \to -dk$

The last row: the flipped limits and the minus on dk **cancel each other**, so the integral is unchanged. That works only because the range is symmetric. k is a dummy variable — like the i in Σ_i — so renaming it changes nothing: $\int_{-\infty}^{\infty} f(k)\, dk = \int_{-\infty}^{\infty} f(-k)\, dk$.

This is the one place $\omega(-k) = \omega(k)$ is used. Without it the exponents would not match either, and the comparison would fail.

So:

$$u^* = \frac{1}{\sqrt{2\pi}} \int dk \; e^{ikx} \left[C^*(-k)\, e^{+i\omega t} + D^*(-k)\, e^{-i\omega t} \right]$$

Now set $u = u^*$. Both are $\int dk\, e^{ikx}[\ldots]$, so match the two time-pieces **separately**:

piece	from u	from u^*	gives
$e^{-i\omega t}$	$C(k)$	$D^*(-k)$	$C(k) = D^*(-k)$
$e^{+i\omega t}$	$D(k)$	$C^*(-k)$	$D(k) = C^*(-k)$

The two rows say the same thing. This is the **same move** as matching the $\hat{x}$ and $\hat{y}$ parts separately — two independent pieces, so each must balance on its own. Note the **$-k$**: D is the conjugate of C at the opposite wavenumber.

So D is not thrown away — it is forced. Put it back:

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int dk \; e^{ikx} \left[C(k)\, e^{-i\omega t} + C^*(-k)\, e^{+i\omega t} \right]$$

Relabel $k \to -k$ in the **second piece only** — it turns into the conjugate of the first piece:

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int dk \; C(k)\, e^{i(kx - \omega t)} + \text{c.c.}$$

Finally write **$A(k) = 2C(k)$**, which is Jackson's amplitude:

FINAL ANSWER — Jackson 7.90

$$u(x,t) = \frac{1}{2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dk \; A(k)\, e^{i[kx - \omega(k)t]} + \text{c.c.}$$

*The "+ c.c." **is** the D-branch, rewritten. A(k) stays a free function — that is what makes the solution general.*

■ THIS IS THE ANSWER — STOP COPYING HERE ■

EXTRA — not part of the answer

1. What A(k) is, if you are given a starting wave (Jackson 7.91).

2nd order in t needs two conditions: u(x,0) and ∂u/∂t(x,0). Put t = 0 in 7.90, then differentiate 7.90 and put t = 0:

$$u(x,0) = \frac{1}{2}\,\frac{1}{\sqrt{2\pi}}\int dk\ [\,A(k) + A^*(-k)\,]\,e^{ikx}$$

$$\frac{\partial u}{\partial t}(x,0) = \frac{1}{2}\,\frac{1}{\sqrt{2\pi}}\int dk\ (-i\omega)[\,A(k) - A^*(-k)\,]\,e^{ikx}$$

Invert both, then add them — $A^*(-k)$ cancels:

$$A(k) = \frac{1}{\sqrt{2\pi}}\int dx\ e^{-ikx}\left[\,u(x,0) + \frac{i}{\omega(k)}\,\frac{\partial u}{\partial t}(x,0)\,\right]$$

Jackson 7.91

2. The lecture uses two conventions — check which one the slide is in.

$$u = \frac{1}{2}\,\frac{1}{\sqrt{2\pi}}\int dk\ A(k)\,e^{i[kx - \omega t]} + \text{c.c.}$$ *identical to Jackson 7.90* (Lec 9, slide 3)

$$u = \frac{1}{\sqrt{2\pi}}\int dk\ A(k)\,e^{i[kx - \omega t]} + \text{c.c.}$$ *no ½ — it sits inside A* (Lec 8, slide 9)

Both give the same u. The amplitudes differ by a factor of 2:

$$A_{\text{lec 8}} = \frac{1}{2}A_{\text{lec 9}} = \frac{1}{2}A_{\text{Jackson}}$$

OPTIONAL **Check.** With $\omega = c|k|$, split at k = 0: k > 0 gives ik(x−ct), k < 0 gives ik(x+ct), so u = f(x−ct) + g(x+ct) — d'Alembert. ✓ Also checked numerically: u came out real and both initial conditions were reproduced to 11 decimals.